



ITSIMPLERA INSTITUTE

Network Administration Internship Program

Week 02 Report

IP Addressing (FLSM & VLSM) and VLAN Configuration

Prepared by: Ghulam Abbas

Role: Network Administration Intern

Supervisor: Jawad Qayum, Senior Network Administrator

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1. What This Week Was About

This week's task was IP addressing and VLANs — two topics that look intimidating on paper but click pretty fast once you're actually running the numbers and building something with them. My goal going in was simple: get comfortable calculating subnets both the fixed way (FLSM) and the efficient way (VLSM), then actually build a small VLAN setup and prove to myself it works, not just copy commands and hope.

By the end of the week, I wanted to be able to explain why a network engineer would choose VLSM over FLSM in a real deployment, and why VLANs matter beyond "the switch config tells you to do it this way."

2. Tools I Used

- Cisco Packet Tracer — for building and configuring the VLAN topology
- GNS3 — to rebuild the same topology and double-check it behaved the same way
- PuTTY — console access into the simulated devices
- Word/Excel — working out and documenting the subnetting problems

3. FLSM vs VLSM — What's the Actual Difference

FLSM is the easy way out, honestly. You pick one subnet size and use it everywhere, so the math is quick and every subnet looks identical on paper. The problem is it wastes addresses if one department needs 5 hosts and another needs 60, FLSM still hands them equal-sized blocks, so a lot of IPs just sit there unused.

VLSM fixes that by sizing each subnet to what it actually needs, starting with whoever needs the most hosts and working down. It takes more effort to calculate by hand, but it's basically how real networks get designed, so it was worth doing properly rather than shortcutting it.

Below are the 10 FLSM problems and 5 VLSM scenarios I worked through, all using Class C (192.168.x.0/24) networks to keep things consistent. For each one I broke the subnet mask down into binary so it's obvious which bits actually got borrowed.

FLSM Worksheet — 10 Problems

FLSM Problem 1

Given Network: 192.168.1.0/24 | **Subnets Required:** 2

Bits Borrowed: 1 New Prefix: /25 New Subnet Mask: 255.255.255.128 Block Size: 128

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.1.0	192.168.1.127	192.168.1.1 – 192.168.1.126	126
2	192.168.1.128	192.168.1.255	192.168.1.129 – 192.168.1.254	126

FLSM Problem 2

Given Network: 192.168.2.0/24 | Subnets Required: 4

Bits Borrowed: 2 New Prefix: /26 New Subnet Mask: 255.255.255.192 Block Size: 64

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.2.0	192.168.2.63	192.168.2.1 – 192.168.2.62	62
2	192.168.2.64	192.168.2.127	192.168.2.65 – 192.168.2.126	62
3	192.168.2.128	192.168.2.191	192.168.2.129 – 192.168.2.190	62
4	192.168.2.192	192.168.2.255	192.168.2.193 – 192.168.2.254	62

FLSM Problem 3

Given Network: 192.168.3.0/24 | Subnets Required: 8

Bits Borrowed: 3 New Prefix: /27 New Subnet Mask: 255.255.255.224 Block Size: 32

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.3.0	192.168.3.31	192.168.3.1 – 192.168.3.30	30
2	192.168.3.32	192.168.3.63	192.168.3.33 – 192.168.3.62	30
3	192.168.3.64	192.168.3.95	192.168.3.65 – 192.168.3.94	30
4	192.168.3.96	192.168.3.127	192.168.3.97 – 192.168.3.126	30
5	192.168.3.128	192.168.3.159	192.168.3.129 – 192.168.3.158	30
6	192.168.3.160	192.168.3.191	192.168.3.161 – 192.168.3.190	30
7	192.168.3.192	192.168.3.223	192.168.3.193 – 192.168.3.222	30
8	192.168.3.224	192.168.3.255	192.168.3.225 – 192.168.3.254	30

FLSM Problem 4

Given Network: 192.168.4.0/24 | Subnets Required: 12

Bits Borrowed: 4 New Prefix: /28 New Subnet Mask: 255.255.255.240 Block Size: 16

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.4.0	192.168.4.15	192.168.4.1 – 192.168.4.14	14
2	192.168.4.16	192.168.4.31	192.168.4.17 – 192.168.4.30	14
3	192.168.4.32	192.168.4.47	192.168.4.33 – 192.168.4.46	14
4	192.168.4.48	192.168.4.63	192.168.4.49 – 192.168.4.62	14
5	192.168.4.64	192.168.4.79	192.168.4.65 – 192.168.4.78	14
6	192.168.4.80	192.168.4.95	192.168.4.81 – 192.168.4.94	14
7	192.168.4.96	192.168.4.111	192.168.4.97 – 192.168.4.110	14
8	192.168.4.112	192.168.4.127	192.168.4.113 – 192.168.4.126	14
9	192.168.4.128	192.168.4.143	192.168.4.129 – 192.168.4.142	14

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
10	192.168.4.144	192.168.4.159	192.168.4.145 – 192.168.4.158	14
11	192.168.4.160	192.168.4.175	192.168.4.161 – 192.168.4.174	14
12	192.168.4.176	192.168.4.191	192.168.4.177 – 192.168.4.190	14

FLSM Problem 5

Given Network: 192.168.5.0/24 | Subnets Required: 4

Bits Borrowed: 2 New Prefix: /26 New Subnet Mask: 255.255.255.192 Block Size: 64

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.5.0	192.168.5.63	192.168.5.1 – 192.168.5.62	62
2	192.168.5.64	192.168.5.127	192.168.5.65 – 192.168.5.126	62
3	192.168.5.128	192.168.5.191	192.168.5.129 – 192.168.5.190	62
4	192.168.5.192	192.168.5.255	192.168.5.193 – 192.168.5.254	62

FLSM Problem 6

Given Network: 192.168.6.0/24 | Subnets Required: 2

Bits Borrowed: 1 New Prefix: /25 New Subnet Mask: 255.255.255.128 Block Size: 128

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.6.0	192.168.6.127	192.168.6.1 – 192.168.6.126	126
2	192.168.6.128	192.168.6.255	192.168.6.129 – 192.168.6.254	126

FLSM Problem 7

Given Network: 192.168.7.0/24 | Subnets Required: 8

Bits Borrowed: 3 New Prefix: /27 New Subnet Mask: 255.255.255.224 Block Size: 32

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.7.0	192.168.7.31	192.168.7.1 – 192.168.7.30	30
2	192.168.7.32	192.168.7.63	192.168.7.33 – 192.168.7.62	30
3	192.168.7.64	192.168.7.95	192.168.7.65 – 192.168.7.94	30
4	192.168.7.96	192.168.7.127	192.168.7.97 – 192.168.7.126	30
5	192.168.7.128	192.168.7.159	192.168.7.129 – 192.168.7.158	30
6	192.168.7.160	192.168.7.191	192.168.7.161 – 192.168.7.190	30
7	192.168.7.192	192.168.7.223	192.168.7.193 – 192.168.7.222	30
8	192.168.7.224	192.168.7.255	192.168.7.225 – 192.168.7.254	30

FLSM Problem 8

Given Network: 192.168.8.0/24 | Subnets Required: 16

Bits Borrowed: 4 New Prefix: /28 New Subnet Mask: 255.255.255.240 Block Size: 16

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.8.0	192.168.8.15	192.168.8.1 – 192.168.8.14	14
2	192.168.8.16	192.168.8.31	192.168.8.17 – 192.168.8.30	14
3	192.168.8.32	192.168.8.47	192.168.8.33 – 192.168.8.46	14
4	192.168.8.48	192.168.8.63	192.168.8.49 – 192.168.8.62	14
5	192.168.8.64	192.168.8.79	192.168.8.65 – 192.168.8.78	14
6	192.168.8.80	192.168.8.95	192.168.8.81 – 192.168.8.94	14
7	192.168.8.96	192.168.8.111	192.168.8.97 – 192.168.8.110	14
8	192.168.8.112	192.168.8.127	192.168.8.113 – 192.168.8.126	14
9	192.168.8.128	192.168.8.143	192.168.8.129 – 192.168.8.142	14
10	192.168.8.144	192.168.8.159	192.168.8.145 – 192.168.8.158	14
16	192.168.8.240	192.168.8.255	192.168.8.241 – 192.168.8.254	14

FLSM Problem 9

Given Network: 192.168.9.0/24 | Subnets Required: 4

Bits Borrowed: 2 New Prefix: /26 New Subnet Mask: 255.255.255.192 Block Size: 64

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.9.0	192.168.9.63	192.168.9.1 – 192.168.9.62	62
2	192.168.9.64	192.168.9.127	192.168.9.65 – 192.168.9.126	62
3	192.168.9.128	192.168.9.191	192.168.9.129 – 192.168.9.190	62
4	192.168.9.192	192.168.9.255	192.168.9.193 – 192.168.9.254	62

FLSM Problem 10

Given Network: 192.168.10.0/24 | Subnets Required: 6 (rounded up to 8)

Bits Borrowed: 3 New Prefix: /27 New Subnet Mask: 255.255.255.224 Block Size: 32

Only 6 subnets were actually needed here, but FLSM has to use a power of 2, so I ended up with 2 spare subnets — not wasted exactly, just held in reserve.

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
1	192.168.10.0	192.168.10.31	192.168.10.1 – 192.168.10.30	30
2	192.168.10.32	192.168.10.63	192.168.10.33 – 192.168.10.62	30
3	192.168.10.64	192.168.10.95	192.168.10.65 – 192.168.10.94	30
4	192.168.10.96	192.168.10.127	192.168.10.97 – 192.168.10.126	30
5	192.168.10.128	192.168.10.159	192.168.10.129 – 192.168.10.158	30
6	192.168.10.160	192.168.10.191	192.168.10.161 – 192.168.10.190	30
7 (spare)	192.168.10.192	192.168.10.223	192.168.10.193 – 192.168.10.222	30

Subnet	Network Address	Broadcast Address	Usable Host Range	# Usable Hosts
8 (spare)	192.168.10.224	192.168.10.255	192.168.10.225 – 192.168.10.254	30

VLSM Worksheet 5 Scenarios

Same idea as above, but now each department gets a block sized to what it actually needs instead of one-size-fits-all. I sorted every scenario largest-to-smallest before allocating, which is the standard way to avoid fragmenting the address space.

VLSM Scenario 1: Small Company Network

Given Network: 192.168.1.0/24

Department	Hosts Needed	Subnet Assigned	Subnet Mask	Broadcast	Usable Range	Usable Hosts
Sales	50	192.168.1.0/26	255.255.255.192	192.168.1.63	192.168.1.1 – 192.168.1.62	62
IT	25	192.168.1.64/27	255.255.255.224	192.168.1.95	192.168.1.65 – 192.168.1.94	30
HR	10	192.168.1.96/28	255.255.255.240	192.168.1.111	192.168.1.97 – 192.168.1.110	14
Guest	5	192.168.1.112/29	255.255.255.248	192.168.1.119	192.168.1.113 – 192.168.1.118	6

Department	Bits Borrowed	Mask (Octet 4, Binary)
Sales	2	11 000000
IT	3	111 00000
HR	4	1111 0000
Guest	5	11111 000

VLSM Scenario 2: Corporate Office with WAN Link

Given Network: 192.168.2.0/24

Department	Hosts Needed	Subnet Assigned	Subnet Mask	Broadcast	Usable Range	Usable Hosts
Engineering	60	192.168.2.0/26	255.255.255.192	192.168.2.63	192.168.2.1 – 192.168.2.62	62
Finance	30	192.168.2.64/27	255.255.255.224	192.168.2.95	192.168.2.65 – 192.168.2.94	30
Reception	12	192.168.2.96/28	255.255.255.240	192.168.2.111	192.168.2.97 – 192.168.2.110	14
Router WAN Link	2	192.168.2.112/30	255.255.255.252	192.168.2.115	192.168.2.113 – 192.168.2.114	2

Department	Bits Borrowed	Mask (Octet 4, Binary)
Engineering	2	11000000
Finance	3	11100000
Reception	4	11110000
Router WAN Link	6	11111100

VLSM Scenario 3: School Network

Given Network: 192.168.3.0/24

Department	Hosts Needed	Subnet Assigned	Subnet Mask	Broadcast	Usable Range	Usable Hosts
Students	100	192.168.3.0/25	255.255.255.128	192.168.3.127	192.168.3.1 – 192.168.3.126	126
Teachers	50	192.168.3.128/26	255.255.255.192	192.168.3.191	192.168.3.129 – 192.168.3.190	62
Admin	20	192.168.3.192/27	255.255.255.224	192.168.3.223	192.168.3.193 – 192.168.3.222	30
Library	10	192.168.3.224/28	255.255.255.240	192.168.3.239	192.168.3.225 – 192.168.3.238	14

Department	Bits Borrowed	Mask (Octet 4, Binary)
Students	1	10000000
Teachers	2	11000000
Admin	3	11100000
Library	4	11110000

VLSM Scenario 4: Hospital Network

Given Network: 192.168.4.0/24

Department	Hosts Needed	Subnet Assigned	Subnet Mask	Broadcast	Usable Range	Usable Hosts
Wards	70	192.168.4.0/25	255.255.255.128	192.168.4.127	192.168.4.1 – 192.168.4.126	126
Labs	30	192.168.4.128/27	255.255.255.224	192.168.4.159	192.168.4.129 – 192.168.4.158	30
Admin	15	192.168.4.160/27	255.255.255.224	192.168.4.191	192.168.4.161 – 192.168.4.190	30
Server Room	6	192.168.4.192/29	255.255.255.248	192.168.4.199	192.168.4.193 – 192.168.4.198	6
Router Link	2	192.168.4.200/30	255.255.255.252	192.168.4.203	192.168.4.201 – 192.168.4.202	2

Department	Bits Borrowed	Mask (Octet 4, Binary)
Wards	1	10000000
Labs	3	11100000
Admin	3	11100000
Server Room	5	11111000
Router Link	6	11111100

VLSM Scenario 5: Bank Branch Network

Given Network: 192.168.5.0/24

Department	Hosts Needed	Subnet Assigned	Subnet Mask	Broadcast	Usable Range	Usable Hosts
Tellers	40	192.168.5.0/26	255.255.255.192	192.168.5.63	192.168.5.1 – 192.168.5.62	62
Loan Dept	20	192.168.5.64/27	255.255.255.224	192.168.5.95	192.168.5.65 – 192.168.5.94	30
Management	10	192.168.5.96/28	255.255.255.240	192.168.5.111	192.168.5.97 – 192.168.5.110	14
Security Cameras	8	192.168.5.112/28	255.255.255.240	192.168.5.127	192.168.5.113 – 192.168.5.126	14
Router Link	2	192.168.5.128/30	255.255.255.252	192.168.5.131	192.168.5.129 – 192.168.5.130	2

Department	Bits Borrowed	Mask (Octet 4, Binary)
Tellers	2	11000000
Loan Dept	3	11100000
Management	4	11110000
Security Cameras	4	11110000
Router Link	6	11111100

Looking at these side by side, the difference is obvious. FLSM leaves either way too much or barely enough room in every subnet, while VLSM tracks pretty closely to what each department actually asked for. That's the whole argument for using it in practice.

4. VLANs — Why They Matter, Not Just How to Configure Them

A VLAN is basically a way of telling a switch “These ports belong together,” even if they're plugged into different switches or sit on different floors. Devices in the same VLAN talk to each other like they're on one big shared switch. Devices in different VLANs can't reach each other at

all unless there's a router somewhere in the picture. It's a simple idea, but it does a lot of heavy lifting in real networks.

Why bother with VLANs at all

- Keeps sensitive traffic (HR, finance, whatever) away from everyone else on the network.
- Cuts down on broadcast noise smaller broadcast domains mean less chatter for every device to process.
- Makes admin easier you group devices by function, not by which cable they happen to be plugged into.
- Scales without rewiring anything adding a new VLAN is a config change, not a trip to the server room.

5. The Topology I Actually Built

I kept this deliberately simple so the VLAN behaviour would be easy to prove: two switches, six PCs, two VLANs, connected by a single trunk link.

Device	Interface	VLAN	IP Address	Subnet Mask
SW1	Fa0/1 (PC0)	10 – Sales	192.168.10.11	255.255.255.0
SW1	Fa0/2 (PC1)	10 – Sales	192.168.10.12	255.255.255.0
SW1	Fa0/3 (PC2)	20 – IT	192.168.10.13	255.255.255.0
SW1	Gi0/1 (Trunk to SW2)	10, 20 (trunk)	—	—
SW2	Fa0/1 (PC3)	10 – Sales	192.168.20.11	255.255.255.0
SW2	Fa0/2 (PC4)	20 – IT	192.168.20.12	255.255.255.0
SW2	Fa0/3 (PC5)	20 – IT	192.168.20.13	255.255.255.0
SW2	Gi0/1 (Trunk to SW1)	10, 20 (trunk)	—	—

The test I was going for: PC0, PC1, and PC2 (all VLAN 10) should ping each other fine across both switches over the trunk. PC3, PC4, and PC5 (VLAN 20) should do the same among themselves. But VLAN 10 shouldn't be able to reach VLAN 20 at all without a router in between that's the part that actually proves segmentation is working, not just that cables are plugged in.

6. Building It — Step by Step

Step 1 – Create the VLANs on SW1

```
enable
configure terminal
vlan 10
name Sales
vlan 20
name IT
exit
```

Step 2 – Assign the access ports on SW1

```
interface range fa0/1 - 2
switchport mode access
switchport access vlan 10
exit
interface fa0/3
switchport mode access
switchport access vlan 20
exit
```

Step 3 – Bring up the trunk link on SW1

```
interface gi0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

Step 4 – Mirror the same setup on SW2

```
enable
configure terminal
vlan 10
name Sales
vlan 20
name IT
exit
interface fa0/1
switchport mode access
switchport access vlan 10
exit
interface range fa0/2 - 3
switchport mode access
switchport access vlan 20
exit
interface gi0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

Step 5 – Set static IPs on each PC

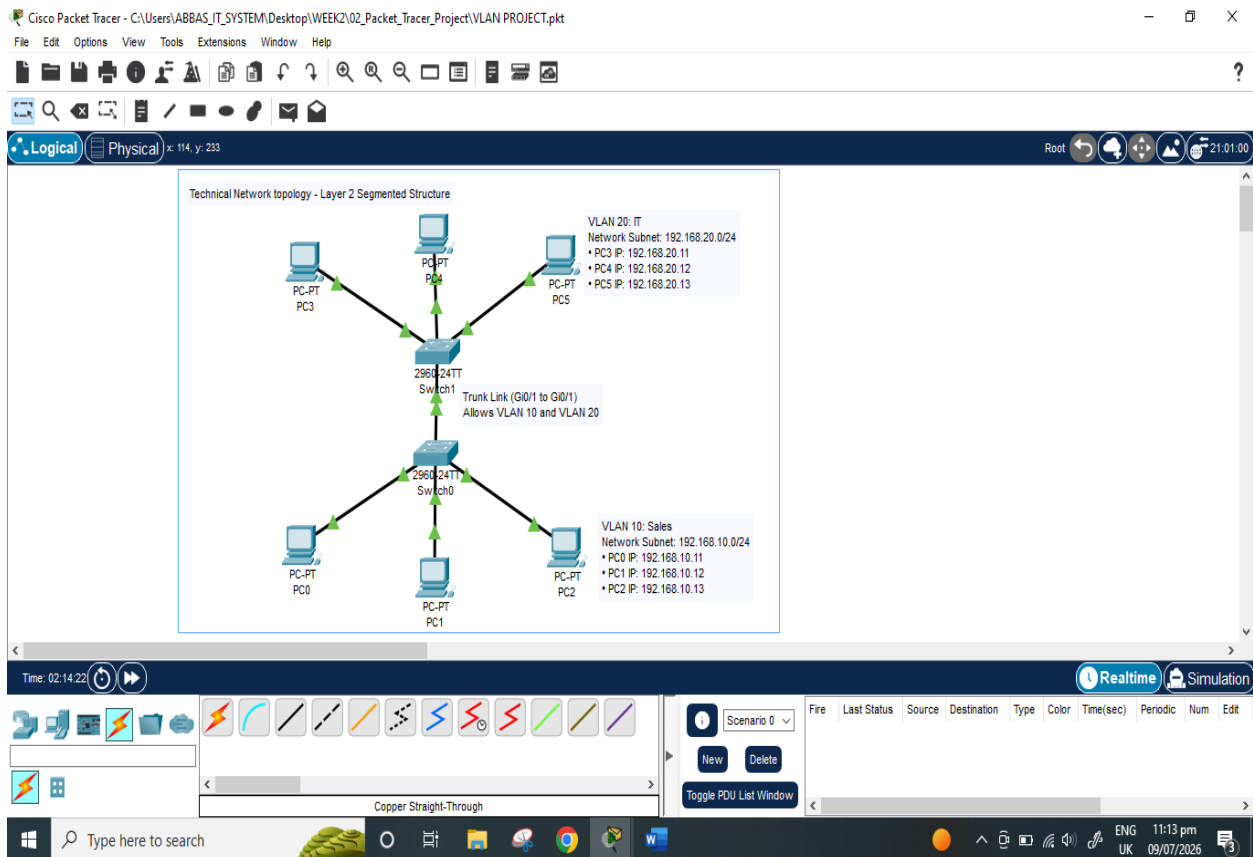
Desktop > IP Configuration on each PC, matching the addressing table above. Gateway left blank for now since there's no router yet.

Step 6 – Save it so a reload doesn't wipe the work

copy running-config startup-config

7. Checking My Work

7.1 Network Topology



7.2 VLAN table

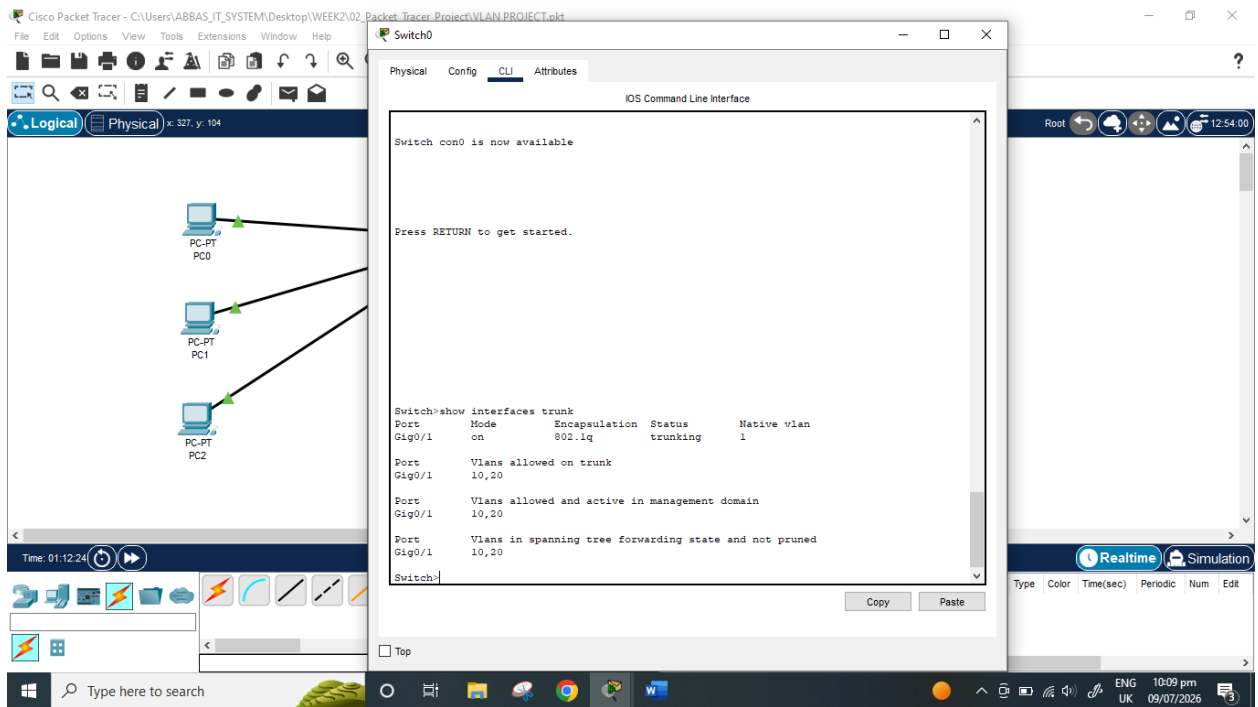
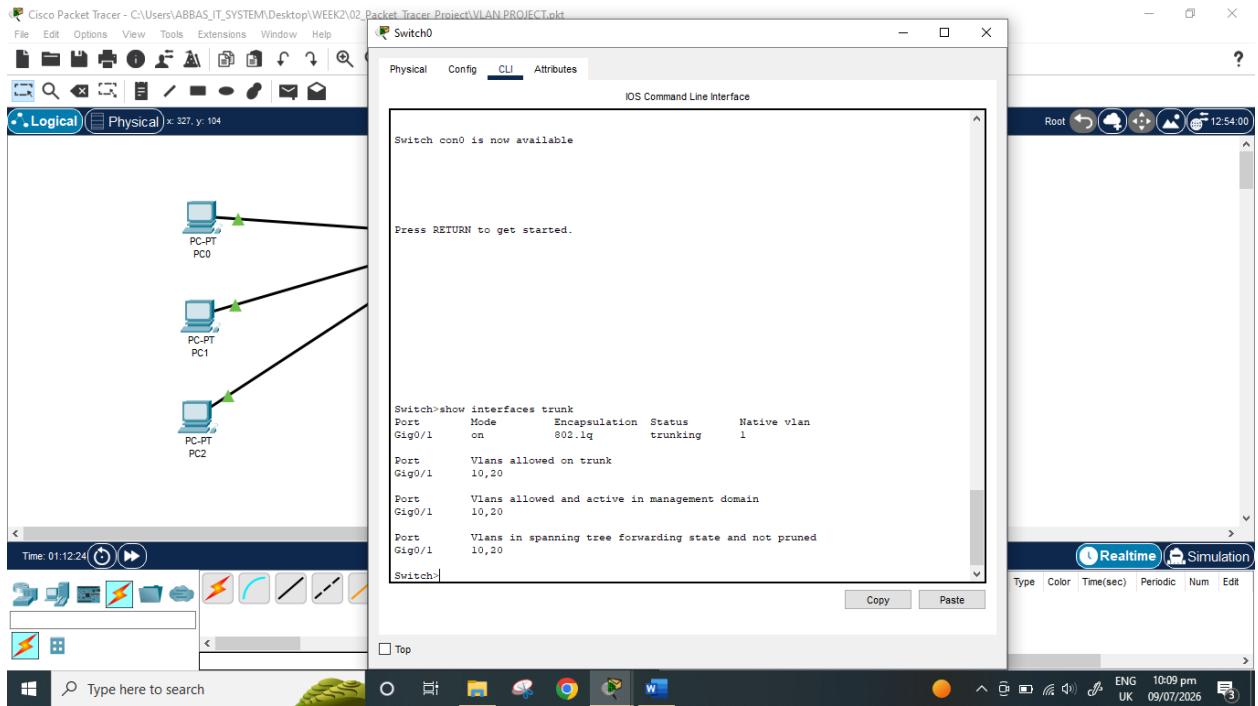
The screenshot displays the Cisco Packet Tracer interface for a network simulation. The main window shows a network topology with two PCs (PC-PT PC3 and PC-PT PC0) connected to a switch (Switch0). The switch's configuration window is open, showing the CLI interface. The configuration includes setting the state of several interfaces (FastEthernet0/1, FastEthernet0/2, FastEthernet0/3, and GigabitEthernet0/1) to 'up'. Below the configuration, the command 'Switch>show vlan brief' is executed, displaying the following table:

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10 Sales	active	Fa0/1, Fa0/2, Fa0/3
20 IT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

The screenshot displays the Cisco Packet Tracer interface for a network simulation. The main window shows a network topology with two PCs (PC-PT PC3 and PC-PT PC0) connected to a switch (Switch1). The switch's configuration window is open, showing the CLI interface. The configuration includes setting the state of several interfaces (FastEthernet0/1, FastEthernet0/2, FastEthernet0/3, and GigabitEthernet0/1) to 'up'. Below the configuration, the command 'Switch>show vlan brief' is executed, displaying the following table:

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10 Sales	active	Fa0/1, Fa0/2, Fa0/3
20 IT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

7.3 Trunk status



7.4 Access port status

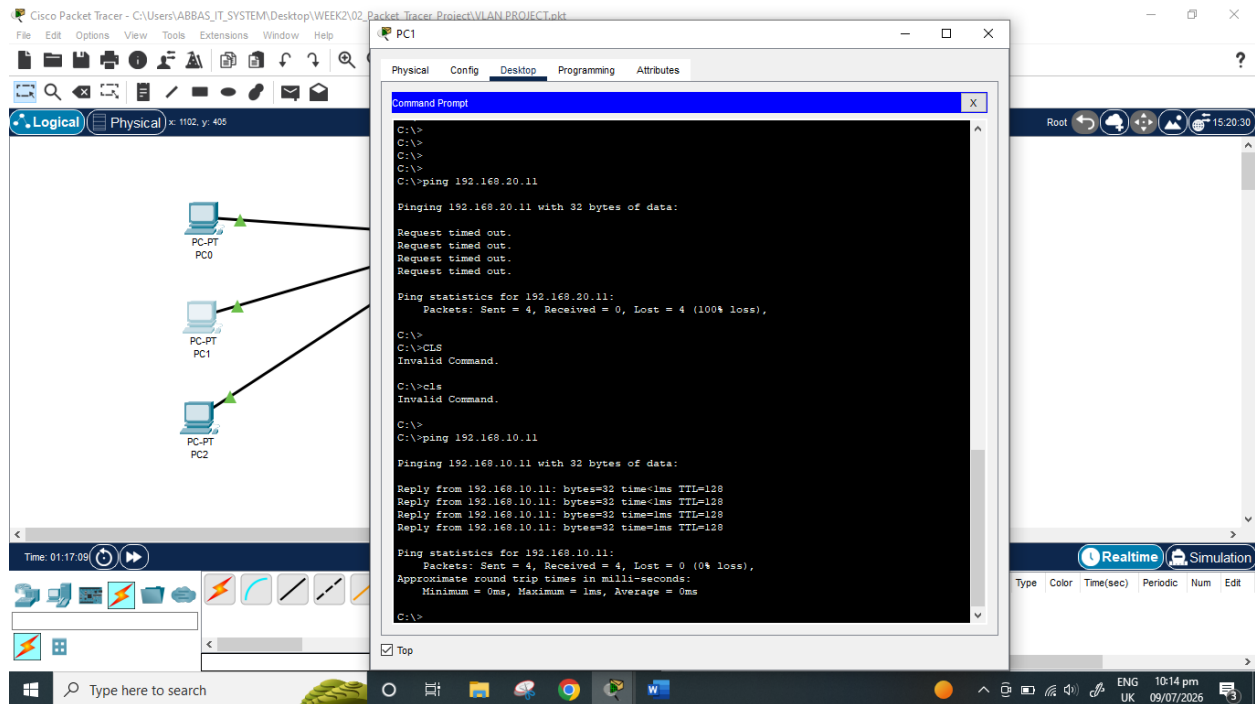
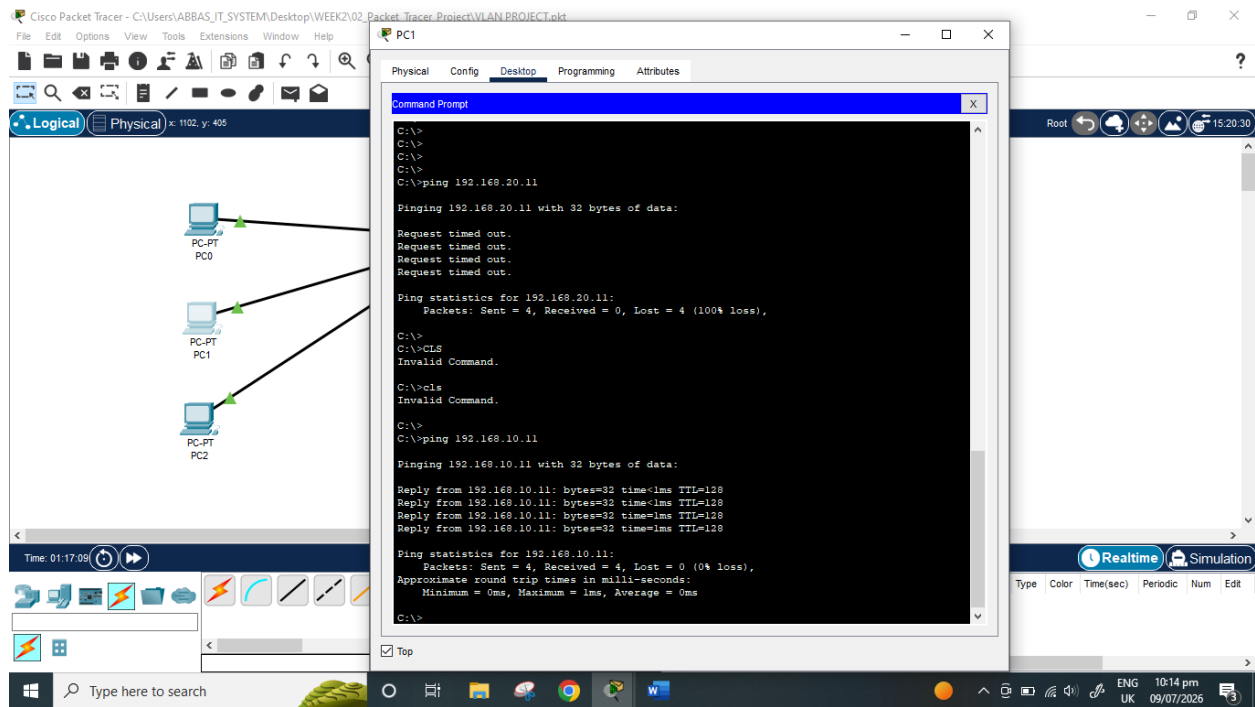
The screenshot shows the Cisco Packet Tracer interface with a network topology on the left and a configuration window for Switch0 on the right. The topology includes two PCs (PC-PT PC3 and PC-PT PC0) connected to a switch. The configuration window displays the 'Physical' tab and the 'CLI' (Command Line Interface) tab. The CLI shows the output of the 'show interface fa0/1 switchport' command, indicating that the interface is in 'static access' mode and is part of VLAN 10 (Sales). The interface is active and has a speed of 1000 Mb/s.

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
1 Default active Fa0/4, Fa0/5, Fa0/6, Fa0/7
Fa0/8, Fa0/9, Fa0/10, Fa0/11
Fa0/12, Fa0/13, Fa0/14, Fa0/15
Fa0/16, Fa0/17, Fa0/18, Fa0/19
Fa0/20, Fa0/21, Fa0/22, Fa0/23
Fa0/24, Gig0/2
Fa0/1, Fa0/2, Fa0/3
10 Sales active
20 IT active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 tnet-default active
Switch>
Switch>show interface fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (Sales)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
--More--
```

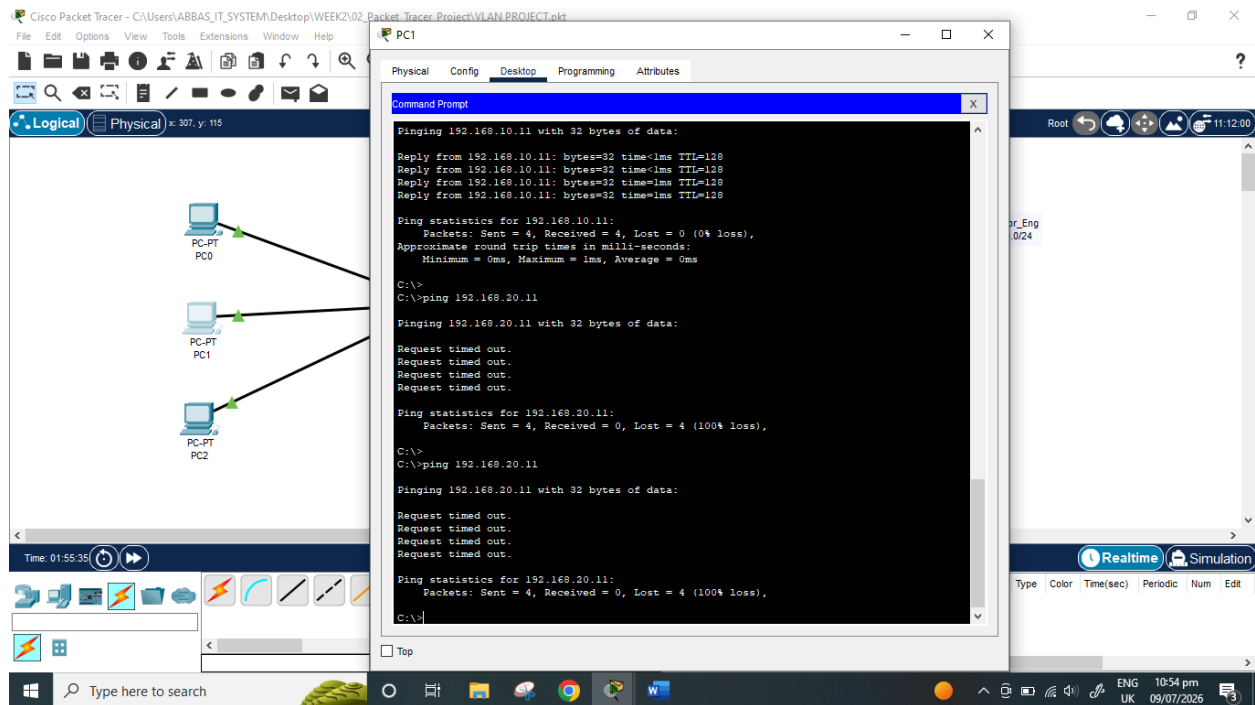
The screenshot shows the Cisco Packet Tracer interface with a network topology on the left and a configuration window for Switch1 on the right. The topology includes two PCs (PC-PT PC3 and PC-PT PC0) connected to a switch. The configuration window displays the 'Physical' tab and the 'CLI' (Command Line Interface) tab. The CLI shows the output of the 'show interfaces trunk' command, indicating that the interface is in 'trunking' mode and is part of VLAN 1. The interface is active and has a speed of 1000 Mb/s.

```
Switch1
Physical Config CLI Attributes
IOS Command Line Interface
Switch>show interfaces trunk
Port Mode Encapsulation Status Native vlan
Gig0/1 on 802.1q trunking 1
Port Vlans allowed on trunk
Gig0/1 10,20
Port Vlans allowed and active in management domain
Gig0/1 10,20
Port Vlans in spanning tree forwarding state and not pruned
Gig0/1 10,20
Switch>show interface f0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 20 (IT)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
--More--
```

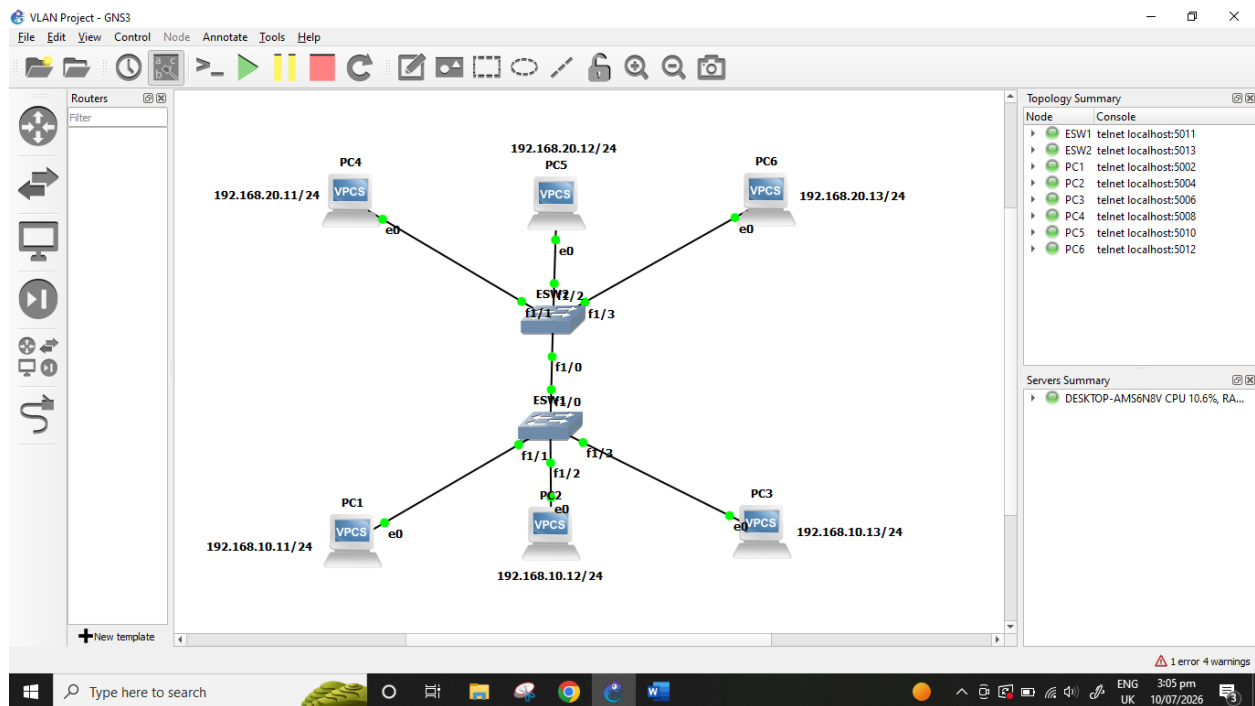
7.5 Same-VLAN ping test (should work)



7.6 Cross-VLAN ping test (should fail — that's the point)



7.7 Network Topology GNS3



7.8 VLAN Implementation

VLAN Project - GNS3

File Edit View Control Node Annotations

Routers

Filter

ESW1

Please update the IOS to enable the "macro" command!

```
ESW1#enable
ESW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW1(config)#vlan 10
% Invalid input detected at '^' marker.
ESW1(config)#exit
ESW1#
*Mar 1 00:06:28.703: %SYS-5-CONFIG_I: Configured from console by console
ESW1#vlan database
ESW1(vlan)#vlan 10 name Sales
VLAN 10 added:
    Name: Sales
ESW1(vlan)#vlan 20 name IT
VLAN 20 added:
    Name: IT
ESW1(vlan)#exit
APPLY completed.
Exiting...
ESW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW1(config)#interface FastEthernet1/1
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#exit
ESW1(config)#interface FastEthernet1/2
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#exit
ESW1(config)#interface FastEthernet1/3
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#exit
ESW1(config)#
```

Topology Summary

Device	Console
ESW1	telnet localhost:5011
ESW2	telnet localhost:5013
PC1	telnet localhost:5002
PC2	telnet localhost:5004
PC3	telnet localhost:5006
PC4	telnet localhost:5008
PC5	telnet localhost:5010
PC6	telnet localhost:5012

Resources Summary

- DESKTOP-AMS6N8V CPU 42.4%, RA...

1 error 4 warnings

ENG UK 2:22 pm 10/07/2026

VLAN Project - GNS3

File Edit View Control Node Annotations

Routers

Filter

ESW2

This is a normal Router with a SN module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```
ESW2#vlan database
ESW2(vlan)#vlan 10 name Sales
VLAN 10 added:
    Name: Sales
ESW2(vlan)#vlan 20 name IT
VLAN 20 added:
    Name: IT
ESW2(vlan)#exit
APPLY completed.
Exiting...
ESW2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW2(config)#interface FastEthernet1/1
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 20
ESW2(config-if)#exit
ESW2(config)#interface FastEthernet1/2
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 20
ESW2(config-if)#exit
ESW2(config)#interface FastEthernet1/3
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 20
ESW2(config-if)#exit
ESW2(config)#
```

Topology Summary

Device	Console
ESW1	telnet localhost:5011
ESW2	telnet localhost:5013
PC1	telnet localhost:5002
PC2	telnet localhost:5004
PC3	telnet localhost:5006
PC4	telnet localhost:5008
PC5	telnet localhost:5010
PC6	telnet localhost:5012

Resources Summary

- DESKTOP-AMS6N8V CPU 60.3%, RA...

1 error 4 warnings

ENG UK 2:26 pm 10/07/2026

7.9 Show vlan-switch brief

The screenshot shows the GNS3 console for ESW1. The configuration commands entered are:

```
ESW1(vlan)#exit
APPLY completed.
Exiting...
ESW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW1(config)#interface FastEthernet1/1
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#exit
ESW1(config)#interface FastEthernet1/2
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#exit
ESW1(config)#interface FastEthernet1/3
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#exit
ESW1(config)#exit
ESW1#
*Mar 1 00:12:11.915: %SYS-5-CONFIG_I: Configured from console by console
ESW1#vlan-switch brief
% Invalid input detected at '^' marker.
ESW1#show vlan-switch brief
```

The output of the `show vlan-switch brief` command is as follows:

VLAN Name	Status	Ports
1 default	active	Fa1/0, Fa1/4, Fa1/5, Fa1/6 Fa1/7, Fa1/8, Fa1/9, Fa1/10 Fa1/11, Fa1/12, Fa1/13, Fa1/14 Fa1/15
10 Sales	active	Fa1/1, Fa1/2, Fa1/3
20 IT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

The right sidebar shows the topology summary with the following devices:

- ESW1 telnet localhost:5011
- ESW2 telnet localhost:5013
- PC1 telnet localhost:5002
- PC2 telnet localhost:5004
- PC3 telnet localhost:5006
- PC4 telnet localhost:5008
- PC5 telnet localhost:5010
- PC6 telnet localhost:5012

The screenshot shows the GNS3 console for ESW2. The configuration commands entered are:

```
ESW2(vlan)#vlan 20 name IT
VLAN 20 added:
  Name: IT
ESW2(vlan)#exit
APPLY completed.
Exiting...
ESW2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW2(config)#interface FastEthernet1/1
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 20
ESW2(config-if)#exit
ESW2(config)#interface FastEthernet1/2
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 20
ESW2(config-if)#exit
ESW2(config)#interface FastEthernet1/3
ESW2(config-if)#switchport mode access
ESW2(config-if)#switchport access vlan 20
ESW2(config-if)#exit
ESW2(config)#exit
ESW2#
*Mar 1 00:07:06.571: %SYS-5-CONFIG_I: Configured from console by console
ESW2#
ESW2#show vlan-switch brief
```

The output of the `show vlan-switch brief` command is as follows:

VLAN Name	Status	Ports
1 default	active	Fa1/0, Fa1/4, Fa1/5, Fa1/6 Fa1/7, Fa1/8, Fa1/9, Fa1/10 Fa1/11, Fa1/12, Fa1/13, Fa1/14 Fa1/15
10 Sales	active	Fa1/1, Fa1/2, Fa1/3
20 IT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

The right sidebar shows the topology summary with the following devices:

- ESW1 telnet localhost:5011
- ESW2 telnet localhost:5013
- PC1 telnet localhost:5002
- PC2 telnet localhost:5004
- PC3 telnet localhost:5006
- PC4 telnet localhost:5008
- PC5 telnet localhost:5010
- PC6 telnet localhost:5012

7.10 Show interface switchport

The screenshot shows the GNS3 interface with a project named 'VLAN Project - GNS3'. The main window displays the configuration for ESW1. The configuration includes enabling the terminal, setting the interface FastEthernet1/0 to trunk mode, and configuring the trunk encapsulation to dot1q. The console output shows the configuration process, including the command 'show interfaces FastEthernet1/0 switchport' and the resulting status of the interface.

```
ESW1#enable
ESW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW1(config)#interface FastEthernet1/0
15 ESW1(config-if)#switchport mode trunk
ESW1(config-if)#swi
*Mar 1 00:01:31.227: %DTP-5-TRUNKPORTON: Port Fa1/0 has become dot1q trunk
ESW1(config-if)#switchport trunk encapsulation dot1q
ESW1(config-if)#no keepalive
ESW1(config-if)#spanning-tree portfast trunk
^
% Invalid input detected at '^' marker.

ESW1(config-if)#show interfaces FastEthernet1/0 switchport
^
% Invalid input detected at '^' marker.

ESW1(config-if)#exit
ESW1(config)#exit
ESW1#enable
*Mar 1 00:03:40.715: %SYS-5-CONFIG_I: Configured from console by console
Name: Fa1/0
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Disabled
Access Mode VLAN: 0 ((Inactive))
Trunking Native Mode VLAN: 1 (default)
Trunking VLANs Enabled: ALL
Trunking VLANs Active: 1,10,20
Priority for untagged frames: 0
Override vlan tag priority: FALSE
Voice VLAN: none
Appliance trust: none
ESW1#
```

The console output also shows the status of the interface:

```
ESW1#show interfaces FastEthernet1/0 switchport
Name: Fa1/0
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Disabled
Access Mode VLAN: 0 ((Inactive))
Trunking Native Mode VLAN: 1 (default)
Trunking VLANs Enabled: ALL
Trunking VLANs Active: 1,10,20
Priority for untagged frames: 0
Override vlan tag priority: FALSE
Voice VLAN: none
Appliance trust: none
```

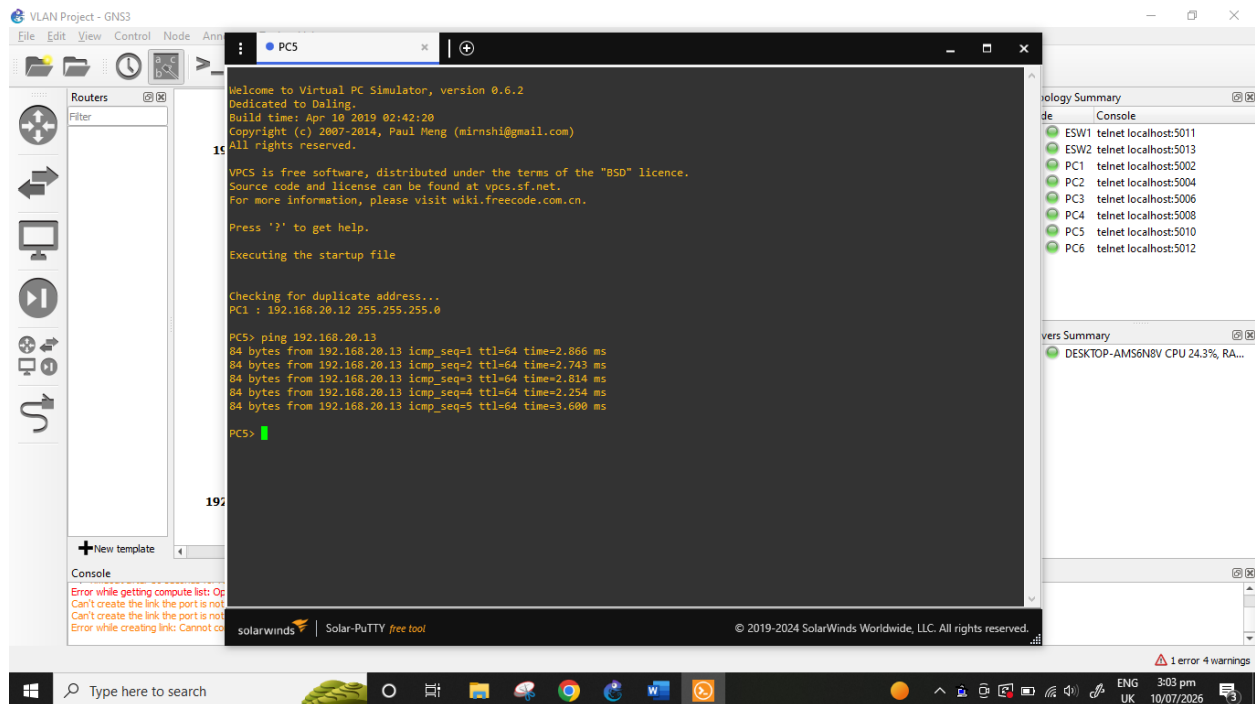
The screenshot shows the GNS3 interface with a project named 'VLAN Project - GNS3'. The main window displays the configuration for ESW2. The configuration includes enabling the terminal, setting the interface FastEthernet1/0 to trunk mode, and configuring the trunk encapsulation to dot1q. The console output shows the configuration process, including the command 'show interfaces FastEthernet1/0 switchport' and the resulting status of the interface.

```
Trunking VLANs Enabled: ALL
Trunking VLANs Active: 1
Priority for untagged frames: 0
Override vlan tag priority: FALSE
Voice VLAN: none
Appliance trust: none
ESW2#enable
ESW2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW2(config)#interface FastEthernet1/0
192.168.20.11/24 ESW2(config-if)#switchport trunk encapsulation dot1q
ESW2(config-if)#switchport mode trunk
ESW2(config-if)#
*Mar 1 00:02:23.015: %DTP-5-TRUNKPORTON: Port Fa1/0 has become dot1q trunk
ESW2(config-if)#no shutdown
ESW2(config-if)#exit
ESW2(config)#exit
ESW2#
*Mar 1 00:04:41.851: %SYS-5-CONFIG_I: Configured from console by console
ESW2#write memory
Building configuration...
[OK]
ESW2#show interfaces FastEthernet1/0 switchport
Name: Fa1/0
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Disabled
Access Mode VLAN: 0 ((Inactive))
Trunking Native Mode VLAN: 1 (default)
Trunking VLANs Enabled: ALL
Trunking VLANs Active: 1,10,20
Priority for untagged frames: 0
Override vlan tag priority: FALSE
Voice VLAN: none
Appliance trust: none
ESW2#
```

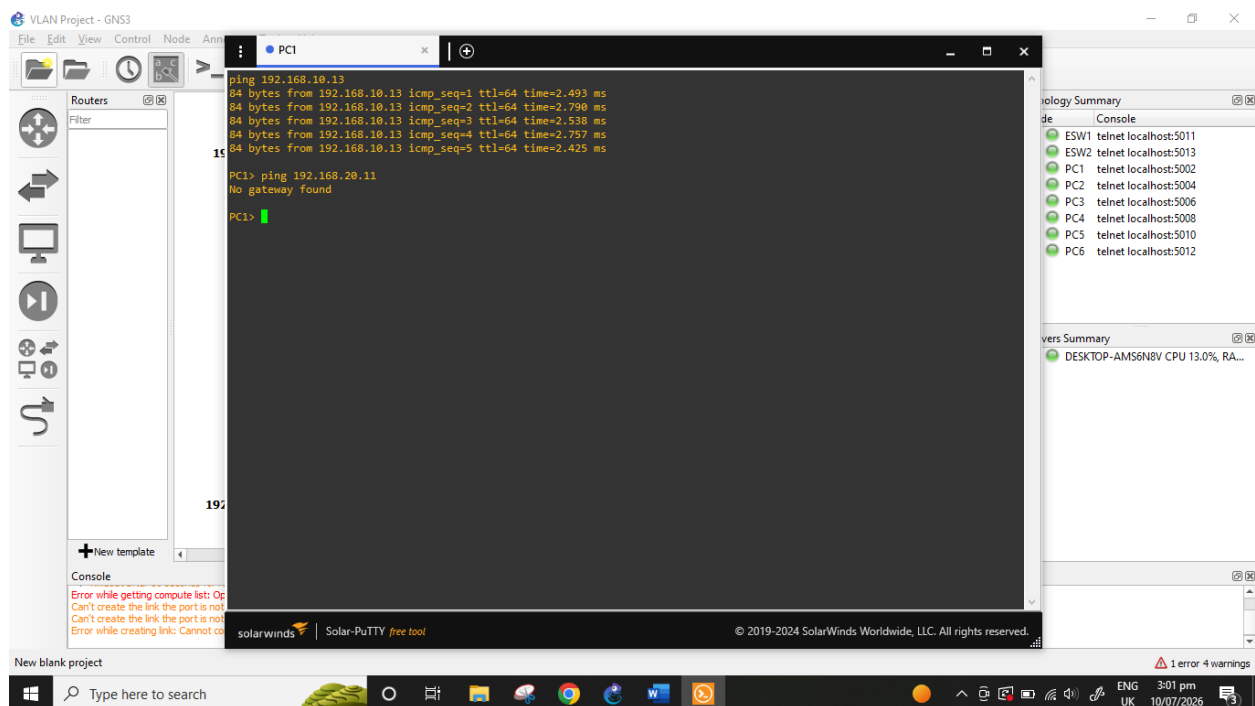
The console output also shows the status of the interface:

```
ESW2#show interfaces FastEthernet1/0 switchport
Name: Fa1/0
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Disabled
Access Mode VLAN: 0 ((Inactive))
Trunking Native Mode VLAN: 1 (default)
Trunking VLANs Enabled: ALL
Trunking VLANs Active: 1,10,20
Priority for untagged frames: 0
Override vlan tag priority: FALSE
Voice VLAN: none
Appliance trust: none
```

7.11 Ping Inter-Vlan



7.12 Inter-Vlan isolation test



“The terminal output above displays the intentional and expected failure of an inter-VLAN communication attempt. While PC1 (192.168.10.11) can ping local hosts within its own broadcast

domain, attempting to ping PC4 (192.168.20.11) in VLAN 20 results in a 'No gateway found' error. Because the source and destination reside on distinct IP subnets, the originating host recognizes that the traffic must be routed via a Layer 3 boundary. Since this topology is explicitly a pure Layer 2 switching architecture with no default gateway or intermediate routing engine (such as a Router-on-a-Stick or a Layer 3 Switch Virtual Interface) configured, the host drops the packet immediately. This terminal response serves as concrete empirical validation that our 802.1Q trunking configuration and database setup are successfully maintaining strict traffic segmentation and security boundaries between the Sales and IT departments.”

Learning Outcomes Achieved

By completing this task, I was able to:

- Explain the difference between FLSM and VLSM, and justify why VLSM is the more realistic approach for uneven host requirements (demonstrated across 10 FLSM problems and 5 VLSM scenarios).
- Calculate network addresses, subnet masks, broadcast addresses, and usable host ranges for both fixed and variable-length subnetting, including binary bit-borrowing breakdowns for each case.
- Design an efficient IP addressing scheme for a multi-department network, correctly sizing each subnet to its actual host count rather than defaulting to equal blocks.
- Explain the purpose of VLANs traffic isolation, reduced broadcast domains, and logical grouping independent of physical switch location.
- Configure and verify VLANs on Cisco switches using Packet Tracer, including creating VLANs, assigning access ports, and establishing a trunk link between two switches.
- Configure access ports and trunk ports correctly, including troubleshooting a real trunk misconfiguration (one-sided trunk setup) and a port left in dynamic auto mode.
- Verify configuration using IOS show commands (show vlan brief, show interfaces trunk, show interfaces fa0/1 switchport) rather than assuming a config was correct simply because it ran without errors.
- Apply network segmentation in practice, confirmed by successful same-VLAN pings across a trunk link and a correctly failing cross-VLAN ping.
- Recreate the same topology in GNS3, adapting to a tool without native IOS CLI support on the default switch model, and document that difference transparently.